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Building a Convolutional Neural Network Model Using TensorFlow and Keras

This article explains how to build, train and deploy a convolutional neural network using TensorFlow and Keras. It is directed at students, faculties and researchers interested in the area of deep learning applications using these networks.

Artificial intelligence (AI) is the science of making intelligent computer programs or intelligent machines. In AI, deep learning (also called deep neural networks) mimics the working of the human brain for processing information and recognising patterns. There are many types of networks available in the wider family of deep neural networks. Among them are convolutional neural networks (CNNs), which are used most commonly for the analysis of visual images. A CNN automates the feature extraction task from raw input images and thus eliminates the requirement of handcrafted feature extraction. This makes it a very promising architecture

for image recognition and classification. A CNN is made of several types of layers, which include convolution and pooling layers, a rectified linear unit (relu) layer, besides the fully connected layer. It also performs two essential operations -- feature learning and classification. Automatic feature learning is performed by a set of the convolution layer, rectified linear unit (relu) layer and pooling layer with batch normalisation, whereas classification is performed by a set of the fully connected layer with activation functions like softmax, sigmoid, etc. Examples of such applications are satellite imagery, video surveillance, leaf disease detection, human pose estimation, etc.